Question	Answer	Marks	Partial Marks
7(b)	107 or 107.2...	**6**	**M5** for [speed =] $\frac{240}{\left(2\times\frac{260}{7}+60\right)}\times 60$ oe OR **B5** for [total time =] 134 or 134.2 to 134.3 or 2.24 or 2.238... or **B4** for (t =) 37.1 or 37.14... OR **M2** for $\frac{t}{60}\times 100+\frac{t+60}{60}\times 110=240$ oe or **M1** for $\frac{t}{60}\times 100$ or $\frac{t+60}{60}\times 110$ oe **M1** for correct equation of form $at = b$ from *their* equation containing two terms in t and involving the speeds. **M1** for $\frac{240}{2\times \textit{their}\, t+60}$ [× 60]
8(a)(i)	37.3 or 37.26 to 37.27	**5**	**M2** for $\pi\times 0.35\times\sqrt{0.35^2+1.5^2}$ oe or **M1** for $0.35^2+1.5^2$ or better **M1** for $\pi\times 0.35^2$ **M1** for $2\times\pi\times 0.35\times 16$
8(a)(ii)	6.35 or 6.349 to 6.351	**3**	**M1** for $\pi\times 0.35^2\times 16$ **M1** for $\frac{1}{3}\times\pi\times 0.35^2\times 1.5$
8(a)(iii)	22.2 or 22.3 or 22.24 to 22.26	**3**	**M2** for 17.5 × 3.5 × 1.4 – 10 × *their***(a)(ii)** or **M1** for 17.5 × 3.5 × 1.4

Question	Answer	Marks	Partial Marks
8(b)	154 or 154.3 to 154.4	**3**	**M2** for $450 \times \left(\sqrt{\frac{98}{200}}\right)^3$ oe or **M1** for $\left(\sqrt{\frac{98}{200}}\right)^3$ or $\left(\sqrt{\frac{200}{98}}\right)^3$ oe or for $\left(\frac{450}{V}\right)^2 = \left(\frac{200}{98}\right)^3$ oe
9(a)(i)	0	**1**	
9(a)(ii)	$\frac{1}{7}$ oe	**1**	
9(b)	$\frac{4}{49}$ oe	**2**	**M1** for $\frac{2}{7} \times \frac{2}{7}$
9(c)(i)	$\frac{2}{21}$ oe	**3**	**M2** for $\frac{2}{7} \times \frac{1}{6} + \frac{1}{7} \times \frac{2}{6}$ oe or **M1** for $\frac{2}{7} \times \frac{1}{6}$ or $\frac{1}{7} \times \frac{2}{6}$ oe seen If 0 scored SC1 for $\frac{4}{49}$
9(c)(ii)	$\frac{19}{21}$ oe	**3**	**M2** for $1 - \left(\frac{2}{7} \times \frac{1}{6}\right) - \left(\frac{2}{7} \times \frac{1}{6}\right)$ oe or **M1** for $\left(\frac{2}{7} \times \frac{1}{6}\right) + \left(\frac{2}{7} \times \frac{1}{6}\right)$ oe ALTERNATIVE **M2** for $\frac{1}{7}$ [× 1] × 3 + $\frac{2}{7} \times \frac{5}{6}$ × 2 or **M1** for $\frac{2}{7} \times \frac{5}{6}$ or $\frac{1}{7}$ [× 1] × 3 If 0 scored SC1 for $\frac{38}{49}$
9(d)	3	**2**	**M1** for $\frac{5}{7} \times \frac{4}{6} \times \frac{2\text{or}3}{5}$
10(a)	$7x^6 - 42x^5$ final answer	**2**	**B1** for one correct term $7x^6$ or $42x^5$ or for $7x^6 - 42x^5$ seen and spoiled

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